

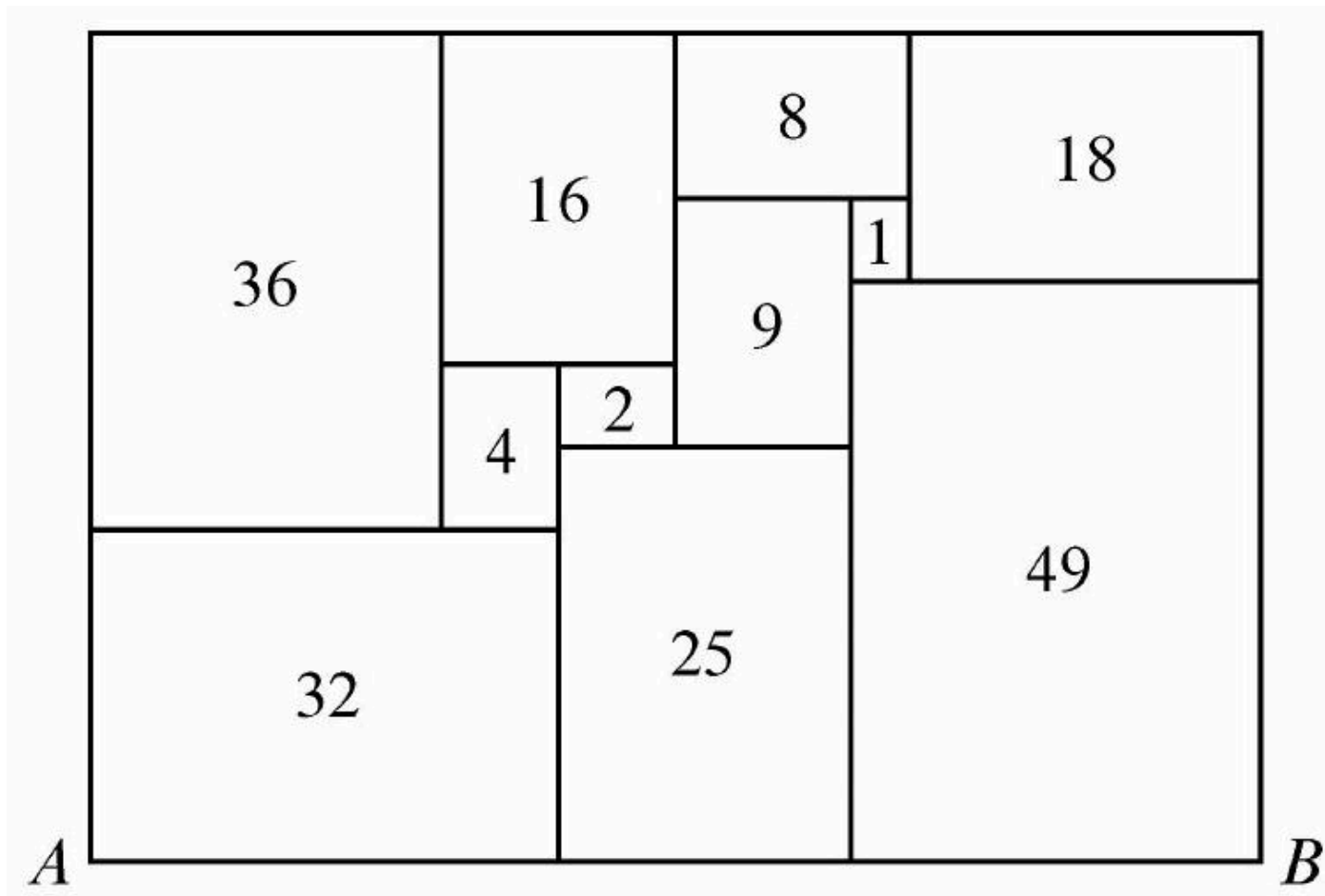
2024 AMC 10A Problem 16 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 16. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 16

Problem:

All of the rectangles in the figure below, which is drawn to scale, are similar to the enclosing rectangle. Each number represents the area of its rectangle. What is length AB ?



Answer Choices:

- A. $4 + 4\sqrt{5}$
- B. $10\sqrt{2}$
- C. $5 + 5\sqrt{5}$
- D. $10\sqrt[4]{8}$
- E. 20

Solution:

Let a represent the length of the shorter side of the rectangle with area 1, and let b represent the length of its longer side. Then $ab = 1$, and the ratio of its long side to its short side is $\frac{b}{a}$, as is true for all of the rectangles. By similarity, the dimensions of the rectangle with area 9 are $3a$ and $3b$, and the dimensions of the rectangle with area 8 are $a\sqrt{8}$ and $b\sqrt{8}$. Because the short sides of the "9" and "1" rectangles add to the long side of the "8" rectangle, $a + 3a = b\sqrt{8}$, and $\frac{b}{a} = \frac{4}{\sqrt{8}} = \sqrt{2}$. The area of the enclosing rectangle is the sum of the numbered areas, which is 200. Therefore $AB \cdot \frac{AB}{\sqrt{2}} = 200$, so

$$AB = \sqrt{200\sqrt{2}} = \sqrt{100\sqrt{8}} = \text{(D)} \boxed{10\sqrt[4]{8}}$$

Note: This dissection was discovered by recreational mathematician Ed Pegg, Jr. The rectangles are similar to standard A4 paper.

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) ↗

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